

Chapter 7

Macroeconomics III
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LIMITS TO
GROWTH? THE
SOLOW MODEL
WITH SCARCE
NATURAL
RESOURCES

INTRODUCTION

- The classical economists, Smith, Ricardo, Malthus, emphasized that land as a production factor is
 - indispensable in the production of aggregate output and
 - in fixed supply.
- They thought that this implied a tendency to negative growth in income per worker in the long run, ending in stagnation and “misery and vice” for the working population.
- At that time agriculture was very important, but “land” is still essential in production – especially in a broader interpretation:
 - All the use we make of nature in human production, e.g., nature’s ability to regenerate clean air and water and to absorb waste products

The classical argument in a nutshell:

- Think of the basic Solow model where the the production function is

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}.$$

Assume that L_t grows at a constant rate, $n > 0$, and that K_t grows at rate n too (as in the model's steady state). Because of constant returns, Y_t will then also grow at rate n . Hence, income per capita, y_t , will be constant, and possibly at a high level.

- In the basic Solow model, a constant y_t can be self-sustaining: the requirement is that capital grows as fast as labour, that is,

$$\begin{aligned} \frac{K_{t+1} - K_t}{K_t} = n &\Leftrightarrow K_{t+1} - K_t = nK_t \Leftrightarrow \\ sY_t - \delta K_t = nK_t &\Leftrightarrow sY_t = (n + \delta)K_t \Leftrightarrow sy_t = (n + \delta)k_t. \end{aligned}$$

In the basic Solow model it **is** possible for savings per capita, sy_t , to cover the replacement investment, $(n + \delta)k_t$, needed to keep capital per capita constant in the long run.

- What if the production function is:

$$Y_t = K_t^\alpha L_t^\beta X^{1-\alpha-\beta},$$

where X is a fixed factor and there are constant returns to K_t, L_t, X ? There will then be decreasing returns to K_t, L_t (since $\alpha + \beta < 1$).

- Assume **hypothetically** that both K_t and L_t grow at the common rate n , so k_t is constant. Then Y_t must grow more slowly than at rate n , i.e., y_t must be decreasing. This is **not** self-sustaining: savings per capita, sy_t , cannot cover the replacement investment, $(n + \delta)k_t$, forever. Hence, in the long run, Y_t must grow even more slowly and y_t has to decrease even faster than under our hypothetical assumption.

- Population growth implies that the increasing input factors press on the limited amount of land. Through diminishing returns to capital and labour, income per capita will decline, ending in “misery and vice” and a halt to population growth.
- Technological progress would be a counteracting factor: there is a race between the tendency to negative growth stemming from land and the tendency to positive growth stemming from technological progress. Who wins?

In this chapter...

- We will set up Solow growth models with
 - natural resources in fixed supply (land) and
 - exhaustible natural resources (oil)to try to answer this question.

THE SOLOW MODEL WITH LAND

- The consumers own the labour, the physical capital and the land (no human capital).
- They rent the productive services of these factors to the firms in competitive markets with rental rates, w_t, r_t and v_t .
- Labour, L_t , and capital, K_t , are accumulated by the consumers as in the Solow model. The stock of land, X , is constant.
- The firms produce output for consumption and investment from (the services of) labour, capital and land.
- Labour augmenting technology, A_t , grows at the exogenous constant rate, g .

THE COMPLETE SOLOW MODEL WITH LAND

$$Y_t = K_t^\alpha (A_t L_t)^\beta X_t^\kappa, \quad \alpha + \beta + \kappa = 1$$
$$K_{t+1} = sY_t + (1 - \delta)K_t, \quad K_0 \text{ given}$$

$$L_{t+1} = (1 + n)L_t, \quad L_0 \text{ given}$$

$$A_{t+1} = (1 + g)A_t, \quad A_0 \text{ given}$$

$$r_t = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} \left(\frac{X}{A_t L_t} \right)^\kappa$$

$$w_t = \beta \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{X}{A_t L_t} \right)^\kappa A_t$$

$$v_t = \kappa \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{X}{A_t L_t} \right)^{\kappa-1}$$

State variables: K_t, L_t, A_t . Constant income shares, e.g., $v_t X / Y_t = \kappa$.

BALANCED GROWTH

- Define $y_t \equiv Y_t / L_t$, $k_t \equiv K_t / L_t$, $x_t \equiv X_t / L_t$.

Then:

$$Y_t = K_t^\alpha (A_t L_t)^\beta X_t^\kappa \Rightarrow y_t = k_t^\alpha A_t^\beta x_t^\kappa.$$

- Defining $g_t^y \equiv \ln y_t - \ln y_{t-1}$, $g_t^A \equiv \ln A_t - \ln A_{t-1} \cong g$ and $g_t^x \equiv \ln x_t - \ln x_{t-1} = -(\ln L_t - \ln L_{t-1}) \cong -n$ one gets:

$$g_t^y \equiv \alpha g_t^k + \beta g_t^A + \kappa g_t^x \cong \alpha g_t^k + \beta g - \kappa n.$$

- What can generate positive/negative growth in y_t ?
- Assume** that the economy converges towards a steady state with balanced growth, involving a constant capital output ratio, $z_t \equiv K_t / Y_t = k_t / y_t$.
- A constant z_t implies $g_t^y = g_t^k$, and hence $(1 - \alpha) g_t^y \cong \beta g - \kappa n$. Hence, under the above assumption g_t^y and g_t^k converge towards:

$$g_t^y \cong \frac{\beta g - \kappa n}{1 - \alpha} = \frac{\beta g - \kappa n}{\beta + \kappa}.$$

CONVERGENCE TO STEADY STATE

- We have already found the growth rate in a hypothetical, stable steady state with balanced growth. But does z_t converge to a constant value, z^* ?
- We analyse the model in terms of z_t . From $y_t = k_t^\alpha A_t^\beta x_t^\kappa$ we get that

$$z_t \equiv k_t / y_t = k_t / (k_t^\alpha A_t^\beta x_t^\kappa) = k_t^{1-\alpha} A_t^{-\beta} x_t^{-\kappa}.$$

Writing this formula for period $t+1$ and doing appropriate substitutions gives:

$$\begin{aligned} z_{t+1} &= k_{t+1}^{1-\alpha} A_{t+1}^{-\beta} x_{t+1}^{-\kappa} = \left(\frac{K_{t+1}}{L_{t+1}} \right)^{1-\alpha} A_{t+1}^{-\beta} \left(\frac{X}{L_{t+1}} \right)^{-\kappa} \\ &= \left(\frac{sY_t + (1-\delta)K_t}{(1+n)L_t} \right)^{1-\alpha} (1+g)^{-\beta} A_t^{-\beta} \left(\frac{X}{(1+n)L_t} \right)^{-\kappa} \Leftrightarrow \end{aligned}$$

$$\begin{aligned}
z_{t+1} &= k_{t+1}^{1-\alpha} A_{t+1}^{-\beta} x_{t+1}^{-\kappa} = \left(\frac{K_{t+1}}{L_{t+1}} \right)^{1-\alpha} A_{t+1}^{-\beta} \left(\frac{X}{L_{t+1}} \right)^{-\kappa} \\
&= \left(\frac{1}{(1+n)(1+g)} \right)^{\beta} \left(sy_t + (1-\delta)k_t \right)^{1-\alpha} A_t^{-\beta} x_t^{-\kappa} \\
&= \left(\frac{1}{(1+n)(1+g)} \right)^{\beta} \left(s \frac{y_t}{k_t} + (1-\delta) \right)^{1-\alpha} k_t^{1-\alpha} A_t^{-\beta} x_t^{-\kappa} \\
&= \left(\frac{1}{(1+n)(1+g)} \right)^{\beta} \left(\frac{s}{z_t} + (1-\delta) \right)^{1-\alpha} z_t \Leftrightarrow \\
z_{t+1} &= \left(\frac{1}{(1+n)(1+g)} \right)^{\beta} \left[s + (1-\delta)z_t \right]^{1-\alpha} z_t^{\alpha}.
\end{aligned}$$

So, the transition curve is:

$$z_{t+1} = \left(\frac{1}{(1+n)(1+g)} \right)^{\beta} \left[s + (1-\delta)z_t \right]^{1-\alpha} z_t^{\alpha}.$$

1. This passes through (0,0).
2. It is everywhere increasing.
3. It has exactly one positive intersection with the 45°-line:

Setting $z_{t+1} = z_t = z$ gives

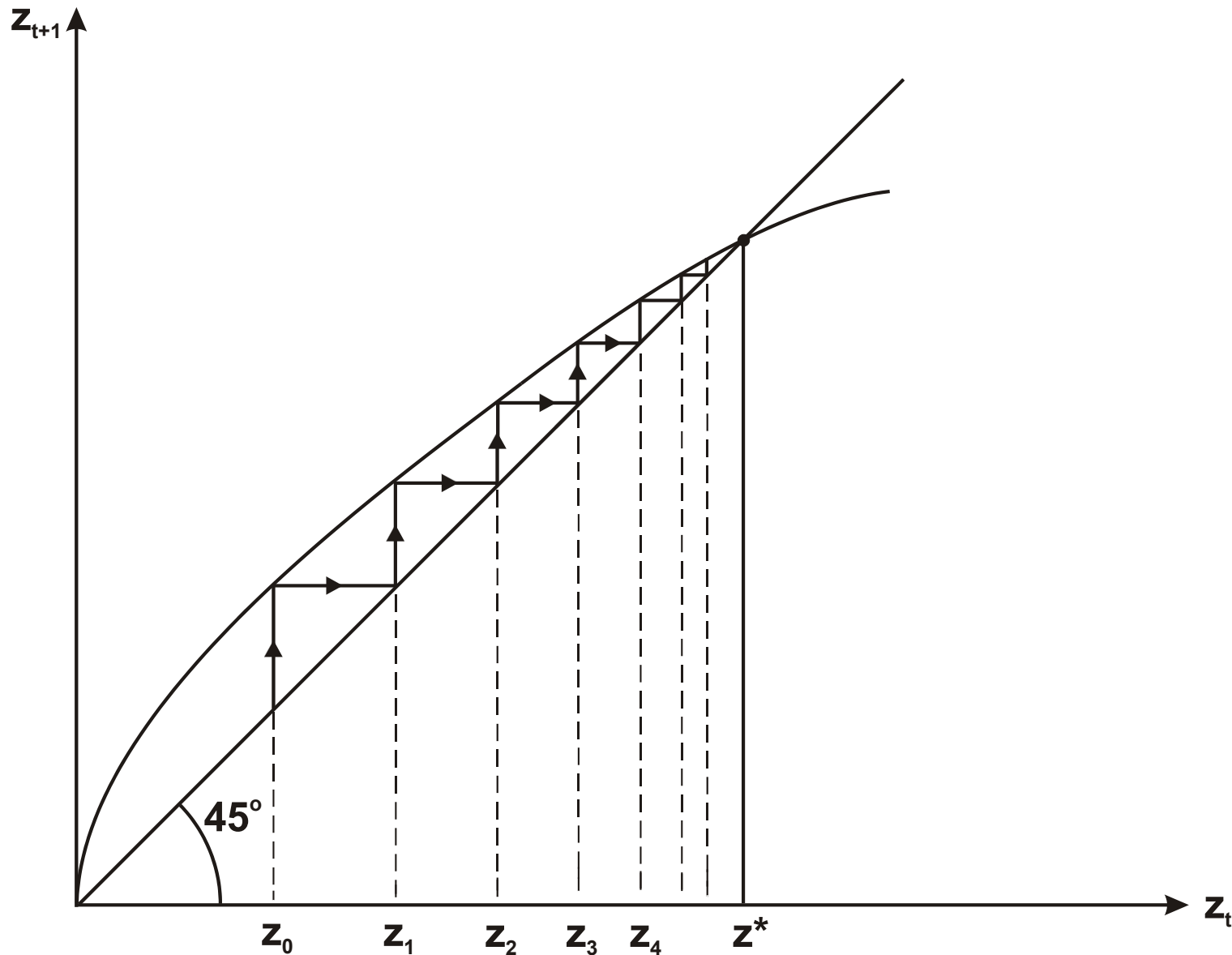
$$z^{1-\alpha} = \left(\frac{1}{(1+n)(1+g)} \right)^{\beta} \left[s + (1-\delta)z \right]^{1-\alpha} \Leftrightarrow$$

$$z = z^* \equiv \frac{s}{\left[(1+n)(1+g) \right]^{\frac{\beta}{\beta+\kappa}} - (1-\delta)} > 0.$$

[Note: setting $\kappa = 0$ gives $z^* = s / (n + g + \delta + ng)$.]

4. The slope at (0,0) is >1 : to show this, find dz_{t+1} / dz_t and evaluate at $z_t = 0$.

With these four properties the transition curve looks as:



There is convergence to a steady state value, z^* .

THE GROWTH RATE IN STEADY STATE

$$g^y = g^k \cong \frac{\beta}{\beta + \kappa} g - \frac{\kappa}{\beta + \kappa} n \Rightarrow (\text{from } g^Y = g^y + n \text{ etc.})$$

$$g^Y = g^K \cong \frac{\beta}{\beta + \kappa} (g + n).$$

Note that $\kappa = 0 \Rightarrow g^y = g$, but $\kappa > 0 \Rightarrow g^y < g$, as long as just $n > -g$. There are two types of growth drag:

1. "Classical case": $g = 0$ and $n > 0$. Negative economic growth, $g^y < 0$. Note that $0 < g^Y = g^K < n$. What happens?

Because of diminishing returns to (K_t, L_t) , as increasing quantities of capital and labour press on the limited amount of land, Y_t cannot keep pace with K_t !

The classics had a further point: from constant income shares, $w_t / y_t = \beta$ and $v_t X / Y_t = \kappa$, it is easy to see that as long as L_t increases and y_t consequently decreases, w_t decreases and v_t increases. Eventually, at sufficiently low w_t , population growth stops ($n = 0$) because of misery and vice among workers. Wages are then at subsistence level, while the rent on land is maximal.

2. "Modern case": $g > 0$ and $n = 0$. There is positive economic growth, but $g^y < g$. Why $g^y < g$? Diminishing returns to $(K_t, A_t L_t)$!
- Which effect is stronger, the negative one from n or the positive from g ? Remember:

$$g^y \cong \frac{\beta}{\beta + \kappa} g - \frac{\kappa}{\beta + \kappa} n.$$

- Inserting plausible parameter values, such as $\beta = 0.6$ and $\alpha = \kappa = 0.2$, gives $g^y \cong (3/4)g - (1/4)n$.
- In that case, $g^y > 0 \Leftrightarrow g > (1/3)n$. In Western economies n is typically between 1/2 and 1 per cent p.a. at the most. Then g just needs to be larger than 1/6 to 1/3 per cent p.a. to create persistent positive growth.
Inserting, e.g., $n = 1\%$ and $g = 2\%$ gives $g^y = 1.75\%$ p.a.
- The classics were right! But in the last two centuries technological progress has kept g up, and birth control has kept n down to levels preventing "misery and vice" in Western countries.

THE BALANCED STEADY STATE GROWTH PATH

- Is it bad to have much land per capita according to our model? No, quite the reverse. Take a look at the per capita production function:

$$y_t = k_t^\alpha A_t^\beta x_t^\kappa$$

A larger $x_t = X_t / L_t$ means more income per worker for given levels of k_t and A_t . Abundance of land does not increase the long-run growth rate, but it does give a higher level of the steady state growth path.

- Dividing by y_t^α on both sides gives: $y_t^{1-\alpha} = z_t^\alpha A_t^\beta x_t^\kappa$. Hence, the steady state growth path is:

$$y_t^* = \left(z^*\right)^{\frac{\alpha}{\beta+\kappa}} A_t^{\frac{\beta}{\beta+\kappa}} \left(\frac{X}{L_t}\right)^{\frac{\kappa}{\beta+\kappa}} = \left(z^*\right)^{\frac{\alpha}{\beta+\kappa}} A_0^{\frac{\beta}{\beta+\kappa}} \left(\frac{X}{L_0}\right)^{\frac{\kappa}{\beta+\kappa}} (1+g)^{\frac{\beta}{\beta+\kappa}t} (1+n)^{\frac{-\kappa}{\beta+\kappa}t}.$$

This is increasing in X / L_0 . It is good to have plenty of land according to this model!

EMPIRICS

$$\ln y_t^* = \frac{\beta}{\beta + \kappa} \ln A_t + \frac{\alpha}{\beta + \kappa} \ln z^* + \frac{\kappa}{\beta + \kappa} \ln \left(\frac{X}{L_t} \right)$$

- Crude approximation: $\ln z^* \cong \ln s - \ln(n + g + \delta) \Rightarrow$

$$\ln y_t^* \cong \frac{\beta}{\beta + \kappa} \ln A_t + \frac{\alpha}{\beta + \kappa} [\ln s - \ln(n + g + \delta)] + \frac{\kappa}{\beta + \kappa} \ln \left(\frac{X}{L_t} \right).$$

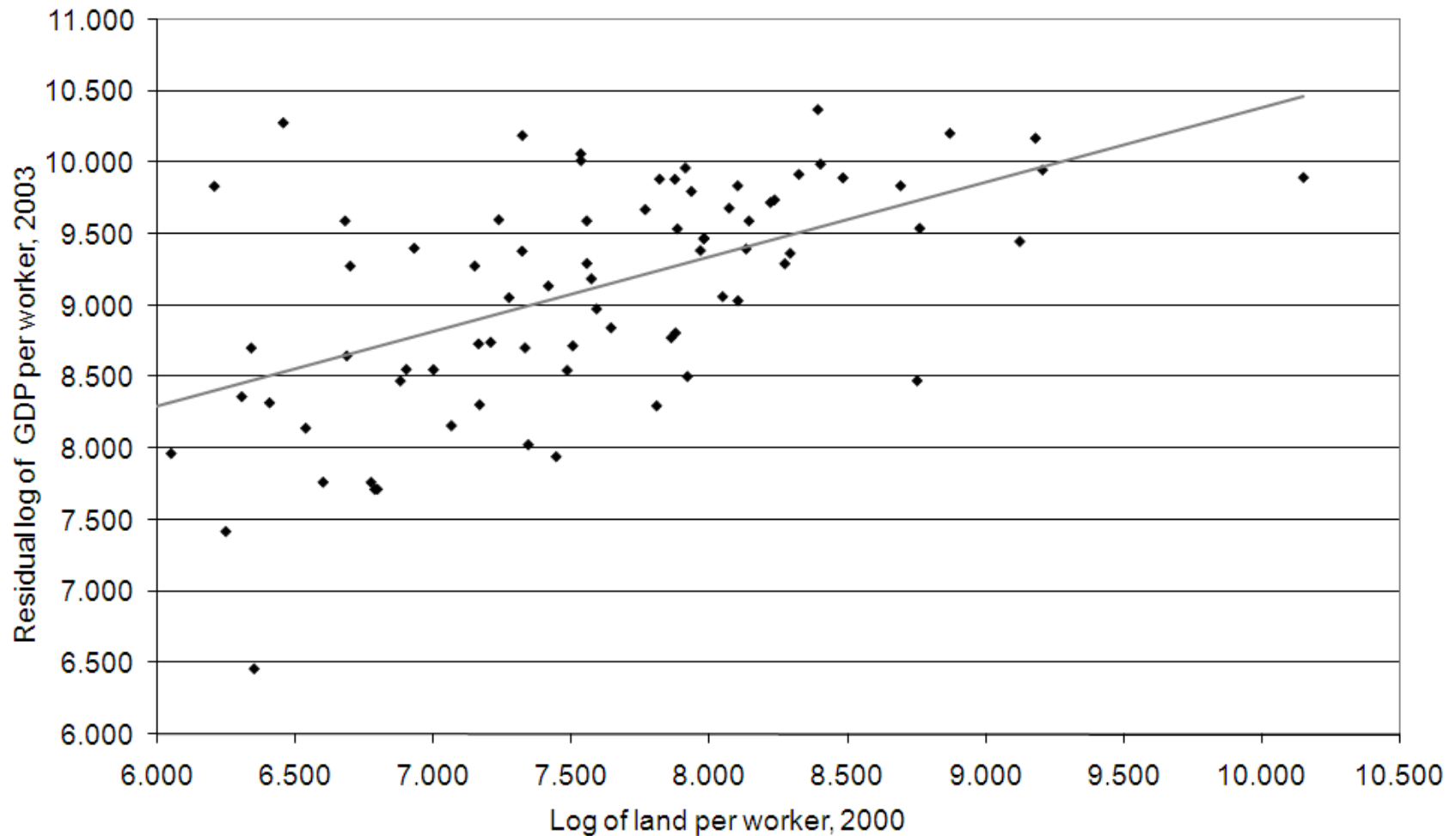
- This suggests the following regression assuming that all countries are in steady state and have the same A_{03} in 2003:

$$\ln y_{03}^i \cong \gamma_0 + \gamma_1 [\ln s^i - \ln(n^i + 0.075)] + \gamma_2 \ln(x_{00}^i).$$

- OLS across 79 countries gives:

$$\ln y_{03}^i \cong 5.16 + \underset{(0.14)}{1.02} [\ln s^i - \ln(n^i + 0.075)] + \underset{(0.11)}{0.52} \ln(x_{00}^i), \text{ adj. } R^2 = 0.61.$$

- Plotting $\ln y_{03}^i - 1.02 \left[\ln s^i - \ln(n^i + 0.075) \right]$ against $\ln(x_{00}^i)$:



THE SOLOW MODEL WITH LAND AND OIL

- Now assume that the consumers also own the remaining stock of energy/oil, R_t , and in period t sell the quantity, E_t of this stock to the firms as energy input. The production function is:

$$Y_t = K_t^\alpha (A_t L_t)^\beta X_t^\kappa E_t^\varepsilon, \quad \alpha + \beta + \kappa + \varepsilon = 1$$

- Capital, labour, technology and land develop in the same way as in the Solow model with land. How about the energy stock? First, it must hold that:

$$R_{t+1} = R_t - E_t$$

- But how large a share of R_t is used as input in each period? We use a simplifying assumption of an exogenous extraction rate, s_E :

$$E_t = s_E R_t, \quad 0 < s_E < \delta.$$

Then $R_{t+1} = (1 - s_E) R_t$, so the growth rate of R_t is $-s_E$, and $R_t = (1 - s_E)^t R_0$.

BALANCED GROWTH

- The per capita production function:

$$Y_t = K_t^\alpha (A_t L_t)^\beta X^\kappa E_t^\varepsilon \Rightarrow$$

$$y_t = k_t^\alpha A_t^\beta \left(\frac{X}{L_t} \right)^\kappa \left(\frac{s_E R_t}{L_t} \right)^\varepsilon = s_E^\varepsilon X^\kappa k_t^\alpha A_t^\beta R_t^\varepsilon L_t^{-\kappa-\varepsilon} \Rightarrow$$

$$g_t^y = \alpha g_t^k + \beta g_t^A + \varepsilon g_t^R - (\kappa + \varepsilon) g_t^L \cong \alpha g_t^k + \beta g - \varepsilon s_E - (\kappa + \varepsilon) n.$$

- If** the economy converges towards a steady state with balanced growth where $z_t = k_t / y_t$ is constant and $g_t^k = g_t^y$, **then** g_t^y must converge towards:

$$g^y = \frac{\beta}{\beta + \kappa + \varepsilon} g - \frac{\kappa + \varepsilon}{\beta + \kappa + \varepsilon} n - \frac{\varepsilon}{\beta + \kappa + \varepsilon} s_E.$$

There **is** in fact convergence towards a steady state with constant z_t .

THE GROWTH RATE UNDER BALANCED GROWTH

$$g^y = \frac{\beta}{\beta + \kappa + \varepsilon} g - \frac{\kappa + \varepsilon}{\beta + \kappa + \varepsilon} n - \frac{\varepsilon}{\beta + \kappa + \varepsilon} s_E.$$

- What is the intuition behind this equation?
Again, there are **diminishing returns** to $(K_t, A_t L_t)$ because increasing amounts of capital and effective labour are set in on limited amounts of natural resources, X and R_t .
- These diminishing returns cause growth drags, i.e., various reductions in the growth rate from g .
- But this time the growth drags are even stronger because R_t is decreasing. This explains the last term.

- Inserting plausible parameter values, $\alpha = 0.2$, $\beta = 0.6$ and $\kappa = \varepsilon = 0.1$ gives:

$$g^y = 0.75g - 0.25n - 0.125s_E$$

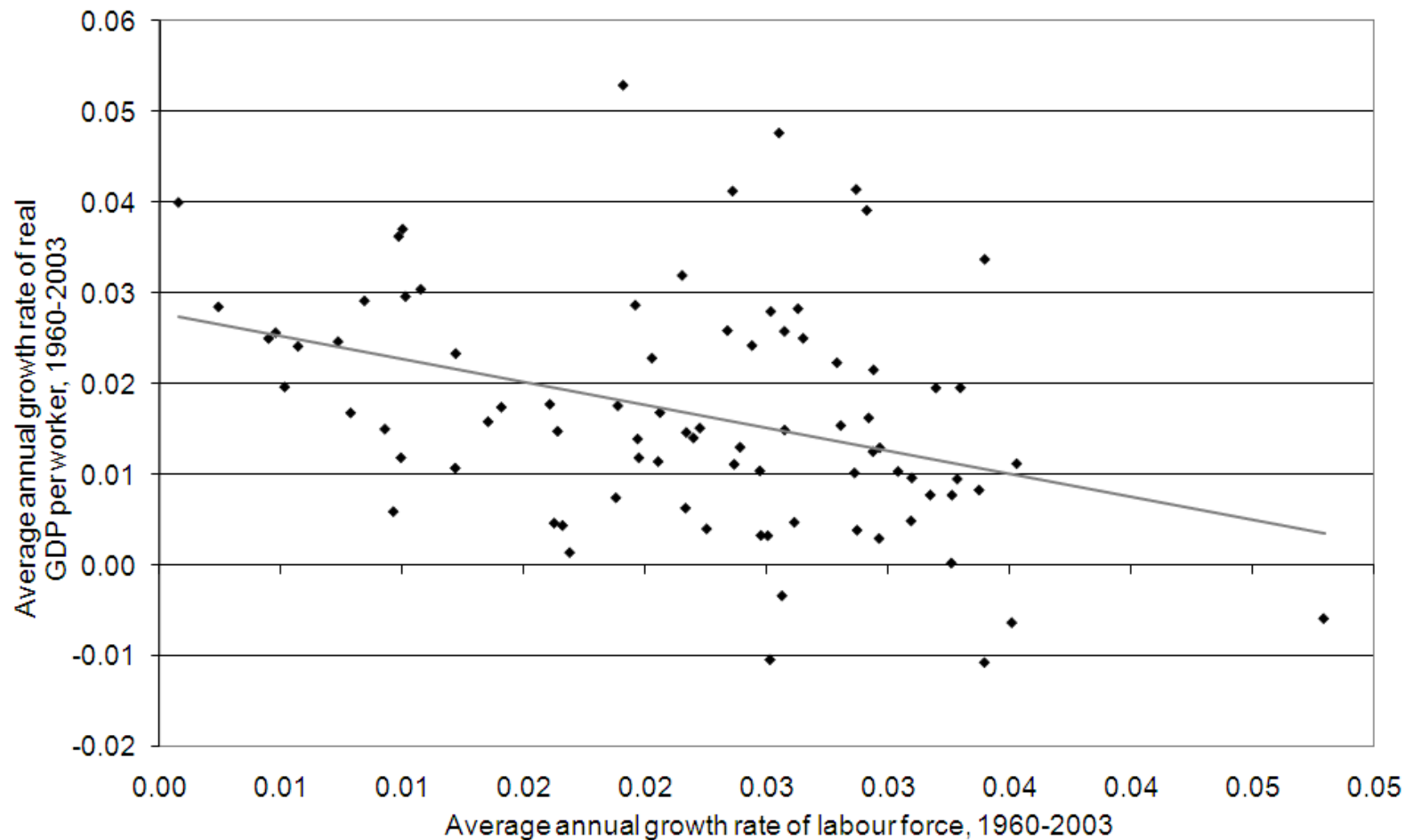
- In a typical Western economy n is at most 0.01 p.a., and an extraction rate $s_E = 0.005$ is often considered reasonable. Combining these gives:

$$g^y = 0.75g - 0.0025 - 0.000625 \cong 0.75g - 0.003.$$

Growth drags: the effect of g is lowered to 3/4 of "full power", and additionally we subtract 0.3 percentage points p.a. If, e.g., $g = 2.4\%$ p.a. we get $g^y = 1.5\%$ p.a. This is growth optimism!

- However, for each percentage point that the population growth rate increases, the growth rate of GDP per worker is lowered by one fourth of a percentage point. Note that if, e.g., $n = 4\%$ p.a. this is a considerable growth drag.

What about empirics?



The slope is around -0.5, but -0.25 is within two standard deviations.

Causality?

CONCLUSION

1. Considering the technological growth rates and population growth rates experienced in Western economies within the last 200 years, **our models suggest** that sustained growth in GDP per capita based on technological progress should be sustainable. This should not be hindered by exhaustion of natural resources or ecological break-downs. But:
2. Considering the population growth rates we find in many developing countries, the classical growth drags cause serious problems. For instance $n = 3\%$ p.a. implies that g has to be larger than one percent p.a. to get positive long-run growth of income per worker at all.

3. The conclusion in 1. is conditioned on continued possibilities of substitution, even when R_t goes to zero. Whether this is a realistic assumption is a central issue in the discussion. Many economists will argue that the assumption of “unlimited possibilities of substitution” expresses the idea that as one type of resource gradually is exhausted, the price of the resource will increase, creating incentives for the discovery of a replacement resource or technology.